

Mayank Wangikar

BS-MS Mathematics, IISER Bhopal

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Research Interests

Functional Analysis, Harmonic Analysis and Operator Algebras.

Education

Indian Institute of Science Education and Research Bhopal, India

BS-MS Dual Degree in Mathematics

Expected Graduation: 2028

CPI: 8.78/10

Research Experience

Summer Research Project – Topological Groups and their Representations

Supervisor: Prof. Prahlad Vaidyanathan

May 2026 – July 2026

Studying topological groups, Haar measure, Baire and Borel structures, representations of compact groups and Peter-Weyl Theorem, and its applications on S^1 and $SU_2(\mathbb{C})$. Reading from Chandrasekharan's *A course on Topological Groups*.

Winter Project – Banach Algebras and C*-Algebras

Supervisor: Prof. Haripada Sau

December 2025

Studied Banach algebras and C*-algebras and their examples with emphasis on weak and weak* topologies, Gelfand Transform, maximal ideals and the Stone-Weierstrass theorem, providing an introduction to operator algebras.

Summer Research Project – Functional Analysis

Supervisor: Prof. Angshuman Bhattacharyya

May-July 2025

Summer group reading project based on MacCluer's *Elementary Functional Analysis*. Topics included normed and Hilbert spaces, bounded operators, Hahn-Banach theorem, Uniform Boundedness Principle, Open Mapping and Closed Graph Theorems, concluding with the spectral theorem for compact self-adjoint operators and presented in weekly discussions.

Reading Project – Approximation Theory

Supervisor: Prof. Prahlad Vaidyanathan

December 2024 – March 2025

Studied normed linear spaces, Banach spaces, compactness and Weierstrass Approximation Theorem and Korovkin's theorem while strengthening the analytical background developed through earlier work in metric spaces.

Summer Research Project – Nonlinear ODEs and Metric Spaces

Supervisor: Prof. Sombuddha Bhattacharyya

May-July 2024

Studied qualitative theory of nonlinear ordinary differential equations, including linearization, Lyapunov stability, Hamiltonian and gradient systems, together with foundational topics in metric spaces such as completeness and compactness.

Relevant courses

Real Analysis, General Topology, Group Theory, Rings and Modules, Fields and Galois Theory, Advanced Linear Algebra, Complex Analysis, Algebraic Topology and Lie Groups and Lie Algebras (Audit).

Awards

- Meritorious Award, SCUDEM IX (SIMIODE), 2024.
 - Selected for the Madhava Nurture Camp (Bhaskaracharya Pratishthana & HBCSE–TIFR), 2024.
 - National Finalist, Inter-Varsity British Parliamentary Debate, IIM Indore, 2023.
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Leadership postions

Chief Editor and Coordinator, Chrysalis (Science Communication Body), IISER Bhopal

Coordinator, Debating Society, IISER Bhopal

Core Committee Member, Mathematics Club, IISER Bhopal

Skills

Python

MATLAB

Mathematica

L^AT_EX

Git